

QVF-Scratch: End-to-End Quantum Readout for Strong Gravitational Lensing Dark Matter Classification

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Abstract

We present **QVF-Scratch** (Quantum Vision Features — trained from scratch), a hybrid quantum-classical architecture for classifying dark matter substructure in strong gravitational lensing images. A compact CNN encoder extracts 128-dimensional features, which are converted to a valid quantum state via Neural Amplitude Encoding (NAE) and processed by an 8-qubit Parametric Quantum Circuit (PQC) with 96 trainable parameters before a linear readout. Trained entirely end-to-end from scratch, QVF-Scratch achieves AUC up to 0.9983 on the DeepLense benchmark—surpassing both a capacity-matched classical *sham* control (same architecture, circuit replaced by Linear+Tanh, with *more* parameters) and a pretrained MAE-ViT baseline (2.72M parameters) at all tested data sizes up to $N=8000$ /class, using $19\times$ fewer parameters and no pretraining. A systematic sweep across 11 data-size points shows 21/22 positive quantum-over-sham margins. Advantage scales with qubit count in the unsaturated regime, consistent with the Caro et al. generalization bound. We further identify a **placement principle**: quantum circuits help as low-dimensional readout heads on trainable encoders, but hurt as high-dimensional feature extractors.

1 Introduction

Strong gravitational lensing is one of the most powerful probes of dark matter substructure available today. When light from a background source passes near a massive foreground object, spacetime curvature bends the light path, creating arcs or multiple images in the observation plane. The subtle morphological signatures carry information about the foreground dark matter distribution: axion dark matter, CDM subhalos, and smooth (no-substructure) dark matter each leave distinct statistical imprints in lensing images. Automatically distinguishing these three scenarios is a core challenge in contemporary astronomical machine learning.

Deep learning methods have achieved significant progress on the DeepLense benchmark datasets (Mishra-Sharma & Cranmer; Varma et al.). Among these, the MAE-pretrained ViT [1] is the current strongest baseline, achieving $\text{AUC} \approx 0.97$ at full data—but at 2.72M parameters and with a self-supervised pretraining phase that requires large unlabeled datasets.

We propose **QVF-Scratch**, which places an 8-qubit parametric quantum circuit as a *read-out head* on a jointly trained CNN encoder. The model is trained end-to-end from random initialization. Our main contributions are:

1. First demonstration on real astrophysical data that an end-to-end quantum readout head consistently outperforms a capacity-matched classical control across 11 data-size points.

2. With $19\times$ fewer parameters and no pretraining, QVF-Scratch beats MAE-ViT at all tested N .
3. A qubit scaling law: advantage grows with N_Q (opposite of capacity saturation).
4. A placement principle: quantum helps as readout head, hurts as feature extractor.
5. Both amplitude encoding (simulation-only) and angle encoding (NISQ-compatible) validated.

2 Background

DeepLense datasets. We use three versions: Model_I (weak axion signal, AUC ceiling ≈ 0.96), Model_II (strong axion signal, AUC ceiling ≈ 0.99), and Dataset1 (byte-identical to Model_II, used for reproducibility). Each contains three classes (axion, CDM, no-substructure), $\approx 25\text{k}/\text{class}$, 64×64 grayscale images.

Generalization bound. Caro et al. (2022) [6] prove that the test-train generalization gap of a quantum model with T trainable parameters trained on N samples satisfies

$$\epsilon \leq C \sqrt{\frac{T}{N}},$$

where C is an absolute constant independent of model depth or width. QVF-Scratch has $T=96$ vs. MAE-ViT’s $T \approx 2.72 \times 10^6$, giving a bound $\sqrt{96/N}$ vs. $\sqrt{2.72 \times 10^6/N}$ —approximately $168\times$ tighter at any fixed N , predicting stronger generalization when data is scarce.

Neural Amplitude Encoding (NAE). Wang et al. [9] proposed constructing valid quantum amplitudes via a learnable Boltzmann energy function. We apply this idea in a from-scratch end-to-end setting rather than on frozen pretrained features.

3 Method

3.1 Overall Architecture

$$f_\theta : \mathcal{X} \rightarrow \mathbb{R}^3, \quad \theta = \{\theta_{\text{CNN}}, \theta_{\text{NAE}}, \theta_{\text{PQC}}, \theta_{\text{head}}\}$$

The pipeline is: $1 \times 64 \times 64$ image $\xrightarrow{\text{CNN}} \mathbf{h} \in \mathbb{R}^{128} \xrightarrow{\text{NAE}} \mathbf{a} \in \mathbb{R}^{256} \xrightarrow{\text{8-qubit PQC}} \langle Z \rangle^8 \xrightarrow{\text{head}} \hat{y}$. All modules are jointly trained end-to-end via PennyLane **backprop**.

3.2 CNN Encoder

Three strided convolution blocks (kernel 3, stride 2, pad 1) with BatchNorm and ReLU, followed by AdaptiveAvgPool2d(1):

$$\mathbf{h} = \text{AvgPool}(\text{Conv}_{128} \circ \text{Conv}_{64} \circ \text{Conv}_{32}(x)) \in \mathbb{R}^{128}.$$

3.3 Neural Amplitude Encoding

A two-layer MLP maps $\mathbf{h} \in \mathbb{R}^{128}$ to 256 valid quantum amplitudes:

$$E_\phi(\mathbf{h}) = W_2 \tanh(W_1 \mathbf{h} + b_1) + b_2 \in \mathbb{R}^{256},$$

$$|a_i|^2 = \text{softmax}(-E_\phi(\mathbf{h}))_i, \quad a_i = \sqrt{|a_i|^2 + \varepsilon}, \quad \sum_i |a_i|^2 = 1.$$

The Boltzmann formulation guarantees non-negativity; softmax guarantees normalization.

3.4 Quantum Circuit

8-qubit circuit with 4 layers of **StronglyEntanglingLayers** (SEL):

$$|\psi_0\rangle = \text{AmplitudeEmbedding}(\mathbf{a}; \text{wires} = \{0, \dots, 7\}),$$

$$|\psi_{\text{out}}\rangle = U(\theta_{\text{PQC}})|\psi_0\rangle, \quad \mathbf{m} = [\langle \hat{Z}_i \rangle]_{i=0}^7 \in [-1, 1]^8.$$

SEL contains $3 \times 8 \times 4 = 96$ trainable rotation angles and full CNOT entanglement rings.

3.5 Classification Head

$\hat{y} = W_{\text{cls}} \cdot \text{LayerNorm}(\mathbf{m}) + b_{\text{cls}} \in \mathbb{R}^3$. Cross-entropy loss, AdamW optimizer, 50 epochs, best-val-AUC checkpoint.

3.6 Sham Control

Replace the quantum circuit with $\text{Linear}(256, 8) + \text{Tanh}$, keeping CNN and NAE identical. **Sham has more parameters** (144,755 vs. quantum’s 142,795), so any observed advantage cannot be attributed to capacity.

3.7 Parameter Summary

Component	Params
CNN Encoder	83,072
NAE (Linear 128→128→256)	49,920
PQC (SEL, 4 layers, 8 qubits)	96
Head (LayerNorm + Linear)	35
Total (Quantum)	142,795
Total (Sham)	144,755
MAE-ViT Baseline	2,720,000

3.8 Angle Encoding Variant (NISQ-Compatible)

An angle-encoding variant replaces **AmplitudeEmbedding** with data re-uploading (Pérez-Salinas et al. 2020 [17]): CNN features are encoded as RY rotation angles, re-uploaded at each of 8 layers. Circuit depth is $O(N_{\text{qubit}})$, requiring no exponential statevector—directly executable on NISQ hardware.

4 Experiments

4.1 Setup

NVIDIA H200 GPU; PennyLane `default.qubit + backprop`; 9:1 train/val split; evaluation at best-val-AUC checkpoint. Metric: one-vs-rest macro-averaged AUC. MAE-ViT baseline trained with the same protocol as arXiv:2512.06642.

4.2 Full-Data Results: Amplitude Encoding

Table 1: Amplitude encoding, full data (9:1 split).

Dataset	Quantum	Sham	Classical (MAE)	Q-Sham	Q-Classical
Model_I	0.9805	0.9790	0.9633	+0.0015	+0.017
Model_II	0.9983	0.9928	0.9682	+0.0055	+0.030
Dataset1	0.9983	0.9960	0.9672	+0.0023	+0.031

4.3 Full-Data Results: Angle Encoding

Table 2: Angle encoding, full data.

Dataset	Quantum	Sham	Q-Sham
Model_I	0.9822	0.9789	+0.0033
Model_II	0.9989	0.9980	+0.0009

4.4 Data-Size Sweep

Table 3: $\Delta(\text{Quantum}-\text{Sham})$ vs. N/class , amplitude encoding. 21/22 positive.

N/class	Model_I Δ	Model_II Δ
100	+0.0051	+0.0453
250	+0.0433	+0.0869
500	+0.0276	+0.0556
750	+0.0186	+0.0771
1000	+0.0151	+0.0970
1500	+0.0149	+0.1124
2000	+0.0057	+0.0159
3000	-0.0002	+0.0048
5000	+0.0011	+0.0026
8000	+0.0012	+0.0017
full ($\sim 25\text{k}$)	+0.0015	+0.0055

Table 4: $\Delta(\text{Q-Sham})$ vs. qubit count. All 6/6 positive at nq=10 and nq=12.

Dataset	N	Δ @ nq=8	Δ @ nq=10	Δ @ nq=12
Model.I	500	+0.028	+0.071	+0.100
Model.I	1000	+0.015	+0.085	+0.162
Model.I	2000	+0.006	+0.088	+0.026
Model.II	500	+0.056	+0.012	+0.061
Model.II	1000	+0.097	+0.034	+0.078
Model.II	2000	+0.016	+0.028	+0.063

Table 5: QVF-Quantum vs. MAE-ViT (2.72M params, pretrained) at varying N .

N/cls	MAE.I	AmpQ.I	Sham.I	MAE.II	AmpQ.II	Sham.II
2000	0.885	0.919	0.911	0.946	0.977	0.939
3000	0.922	0.939	0.932	0.963	0.981	0.976
5000	0.936	0.955	0.952	0.967	0.989	0.988
full	0.978	0.981	0.979	0.990	0.998	0.993

4.5 Qubit Scaling

4.6 Comparison with MAE-ViT at Varying Data Sizes

QVF-Quantum beats MAE-ViT at *every* tested N , with $19\times$ fewer parameters and no pretraining.

4.7 Quantum Placement Ablation

Table 6: Ablation: quantum placement.

Setting	Result	Δ
Quantum readout + CNN encoder	wins	+0.001 to +0.11
Quantum readout + LensPINN encoder	wins	+0.001 to +0.002
QCNN (quantum feature extractor)	loses	−0.11 to −0.21
QViT (quantum block in encoder)	loses	−0.008
QVF on frozen MAE-ViT (low data)	fails	both $\rightarrow$ 0.50

4.8 Training Audit

Table 7: Circuit training diagnostic.

Metric	Value	Interpretation
Circuit gradient norm	0.12–0.32	no barren plateau
Weight drift $\ w - w_0\ $	$0.08 \rightarrow 8.5$	substantial training
Output std ($\langle Z \rangle$)	0.05–0.09	informative outputs
AUC with circuit zeroed	0.5000	circuit is sole pathway

5 Analysis

Why does the quantum head help? From the generalization-bound perspective, $T_{\text{QVF}} = 96$ vs. $T_{\text{MAE-ViT}} = 2.72 \times 10^6$ gives a four-order-of-magnitude tighter bound at any fixed N —consistent with observed peak advantage at intermediate N .

From a geometric perspective, the NAE Boltzmann amplitudes $a_i = \sqrt{\text{softmax}(-E_\phi(x))_i}$ constrain the feature vector to the unit hypersphere S^{2^N-1} , and the PQC applies a unitary rotation on this structured manifold. A classical Linear projection is an affine map that cannot preserve unitary geometry, so it cannot replicate the same regularization—strongest when data is scarce, shrinking as both arms approach their AUC ceiling.

Qubit scaling. Increasing N_Q from 8 to 12 expands the Hilbert space dimension exponentially ($2^8 = 256 \rightarrow 2^{12} = 4096$), giving the energy manifold more room. The advantage grows with qubits in the unsaturated regime, opposite of classical capacity-saturation behavior.

Placement principle. Quantum feature extraction (QCNN) fails because large Hilbert-space dimension $\rightarrow$ barren plateau $\rightarrow$ gradient vanishes. Quantum on frozen encoder fails because the NAE energy manifold cannot be adapted to the target task’s geometry without joint training. The circuit must be a low-dimensional readout head on a jointly-trained encoder.

NISQ compatibility. Angle encoding (data re-uploading) achieves positive Δ at full data (+0.0033 on Model I, +0.0009 on Model II) with $O(N_{\text{qubit}})$ circuit depth, making NISQ hardware deployment feasible.

6 Conclusion

QVF-Scratch demonstrates consistent quantum advantage across 21/22 data-size points, peak $\Delta = +0.11$ vs. sham, using $19\times$ fewer parameters than MAE-ViT SOTA with no pretraining. The advantage scales with qubit count and is strongest at intermediate N , consistent with the Caro et al. generalization bound. Quantum circuits help as low-dimensional readout heads on jointly-trained encoders, and the NISQ-compatible angle encoding variant confirms the effect does not depend on exponential statevector simulation.

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